Weighted dyadic summability does not control spectral-multiplier variation

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Result. The weighted replacement proposed after Theorem 3.3 of Kriegler–Weis fails. For $c = 4$ there is a bounded smooth multiplier m, with $m(0) = 0$, such that

$$\sum_{n\in\mathbb{Z}} \frac{\|m(2^n\cdot)\varphi_0\|_{W_2^4(\mathbb{R})}}{1+|n|} < \infty,$$

but, already for the identity operator on a one-point space, $t \mapsto m(tA)1 = m(t)$ has infinite q-variation for every finite q.

1 Source question

The concluding remarks of [1] ask whether, in the q-variation part of Theorem 3.3 and the time-dependent maximal estimate of Theorem 5.1, the unweighted dyadic Sobolev sum can be relaxed by dividing its n-th term by $1 + |n|$ (possibly with an additional logarithm).

2 Idea of the proof

The denominator $1 + |n|$ makes Sobolev cost cheap at remote dyadic indices. We place the j-th oscillatory block near the log-dyadic index $N_j = 2^{6j}$, but give it only frequency $K_j = 2^j$. Differentiating four times costs $O(K_j^4)$, so the proposed weighted sum is bounded by $C\sum_j K_j^4/N_j = C\sum_j 2^{-2j}$.

Variation measures a different feature. On the core of the j-th block, the cutoff is identically one and the multiplier makes order K_j alternating jumps of fixed size. Thus its q-variation is at least a constant times $K_j^{1/q}$, which diverges. Taking $A = I$ on a one-point space turns the operator conclusion into precisely this scalar variation statement, while satisfying all functional-calculus hypotheses.

3 Construction and proof

Fix a nonzero $\eta \in C_c^\infty((-1/4, 1/4))$ such that $0 \le \eta \le 1$ and $\eta = 1$ on $[-1/8, 1/8]$. Put

$$N_j = 2^{6j}, \qquad K_j = 2^j \quad (j \ge 1),$$

and, for $\lambda > 0$, define

$$m(\lambda) = \sum_{j\ge1} \eta(\log_2\lambda - N_j)\sin\bigl(2\pi K_j(\log_2\lambda - N_j)\bigr), \qquad m(0) = 0.$$

Concluding remarks Already for classical (i.e. without q-variation or supremum in $t > 0$) Hörmander multiplier theorems, a nice description of the exact norm $\|m(A)\|_{L^p \to L^p}$ in terms of a function norm $\|m\|$ of the spectral multiplier is not known today. This problem is equally present for our q-variational and maximal spectral multiplier estimates in this article and only a step by step progression of sufficient conditions in the form $\|t \mapsto m(tA)(\cdot)\|_{L^p \to L^p(V^q)} \lesssim \|m\|_{\dots}$ seems to be manageable. In this direction, it would be interesting to know whether in the contexts of Theorem 3.3 2. (q-variation) and of Theorem 5.1 2. (maximal estimate with time-dependent f_t) of semigroup generators, one can relax the summation condition to

$$\sum_{n \in \mathbb{Z}} \frac{\|m(2^n \cdot)\varphi_0\|_{W_2^c(\mathbb{R})}}{1 + |n|} < \infty$$

as for maximal estimates in the euclidean case [CGHS], or with an additional factor $\log(|n|+2)$ as in [Choi]. As another possible relaxation, one can ask the question, whether

$$\sum_{n \in \mathbb{Z}} \|m(2^n \cdot)\varphi_0\|^2_{W_q^c(\mathbb{R})} < \infty$$

is sufficient for our estimates, which is known to be true for the classical maximal estimate of the euclidean Laplacian [CGHS, (1.3)]. For this, it would be interesting to know whether there are other approaches to q-variational and time dependent maximal estimates than ours, in case of the euclidean Laplacian or sub-Laplacians on Lie groups. Also q-variational estimates for spectral multipliers that do not decay at ∞ are not well understood. Already the scalar case $Y = \mathbb{C}$ would be interesting.

In the context of Section 4, it would be interesting to get information on the norm $C_{p,q,d,Y}$ in Theorem 4.2 and on $C_{p,q,Y}$ in Proposition 4.4, depending on $p, q(, d)$ and Y. Also one could try to find a proof of Proposition 4.4 using Hörmander functional calculus that also covers small, so finally all, dimensions.

8 Acknowledgments

Figure 1: The exact concluding questions in arXiv:2404.01893, PDF page 31.

The log-supports are disjoint, so the sum is locally finite, $m \in C^\infty([0,\infty))$, and $|m| \le 1$.

Theorem 1. *For every fixed dyadic cutoff $\varphi_0 \in C_c^\infty((1/2, 2))$,*

$$\sum_{n \in \mathbb{Z}} \frac{\|m(2^n \cdot)\varphi_0\|_{W_2^4(\mathbb{R})}}{1 + |n|} < \infty, \qquad \|m\|_{V^q(0,\infty)} = \infty \quad (1 \le q < \infty).$$

Proof. On the support of φ_0, the variable $u = \log_2 \lambda$ ranges over a fixed compact interval. Thus $m(2^n\lambda)\varphi_0(\lambda)$ can meet the j-th log-support only when $|n - N_j| \le C_0$, with C_0 independent of j; for each j there are only $O(1)$ such integers n. The chain rule on this fixed compact λ-interval gives

$$\|m(2^n \cdot)\varphi_0\|_{W_2^4} \le C(1 + K_j)^4 \le C' K_j^4$$

for those n. Since $1 + |n| \ge c N_j$ there,

$$\sum_n \frac{\|m(2^n \cdot)\varphi_0\|_{W_2^4}}{1 + |n|} \le C \sum_{j \ge 1} \frac{K_j^4}{N_j} = C \sum_{j \ge 1} 2^{-2j} < \infty.$$

On the core interval $|\log_2 t - N_j| \le 1/8$, the cutoff equals one. Sampling successively at a maximum and the following minimum of the sine wave produces at least $K_j/4 - 2$ increments of magnitude 2. Hence

$$\|m\|_{V^q} \ge 2(K_j/4 - 2)^{1/q}.$$

As $K_j \to \infty$, this proves infinite q-variation. $\square$

4 Operator hypotheses and scope

Let both underlying measure spaces be one point, let $p = 2$, $Y = \mathbb{C}$, and let $A = I$. Then A is 0-sectorial, has a bounded Hörmander calculus of every admissible order, and $e^{-tA} = e^{-t}I$ is positive and contractive. Taking $\alpha = 1$, the choice $c = 4$ satisfies the smoothness inequality in Theorem 3.3. For $f = 1$, however, $m(tA)f = m(t)$, so the asserted variational conclusion fails.

This disproves the proposed relaxation in the q-variation context and therefore gives a negative answer to the blanket question. It does not settle whether the analogous weighted assumption might suffice for the time-dependent maximal estimate, nor the separate square-summability proposal. The optional extra factor $\log(|n| + 2)$ in the numerator is also harmless here, since $\sum_j j\, 2^{-2j} < \infty$.

References

[1] C. Kriegler and L. Weis, *q-variational Hörmander functional calculus and Schrödinger and wave maximal estimates*, arXiv:2404.01893.